

Relational Database Design

Intro Join Dependencies

Uni Auckland Databases 51

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Database Joins

- ▶ Arguably a signature operation of relational databases.
- ▶ Perceived reasons why Joins are Underutilized
 - ▶ **Misconception:** Joins are complex and slow.
 - ▶ **Reality:** Properly indexed joins are efficient.
 - ▶ **Example:** Combining customer and order data.

Implications of Not Doing Joins

含义

- ▶ Data redundancy.
- ▶ Increased storage requirements.
- ▶ increased processing time and resources.
- ▶ More low-level code.
- ▶ Bugs.
- ▶ Inconsistency.

Relation Notation 符号

- ▶ Using relation notation:
`person(person_id*, surname, givenname)`
- ▶ `person_id`, `surname` and `givenname` are attributes
- ▶ `person_id` is a candidate key.
- ▶ Three concepts: the attribute set (aka schema), the relation as updateable object, a concrete state of that.
- ▶ `person({person_id, surname, givenname})`

Natural Joins

- ▶ Our goal: use Natural joins to the maximum extent
- ▶ Natural joins combine tables based on common attributes
- ▶ Example:
 - ▶ `person(person_id*, surname, givenname)`
 - ▶ `address(person_id*, street, city)`
 - ▶ Natural join: `person ⋈ address`
- ▶ Example Schema
 - ▶ `person(person_id*, surname, givenname)`
 - ▶ `address(person_id, street, city)`
 - ▶ `contact(person_id*, phone, email)`
- ▶ Natural joins:
 - ▶ `person ⋈ address`
 - ▶ `person ⋈ contact`

可能 - 个人名字
地址

Decompositions

- ▶ Decomposition of a schema R into schemas $R_1, R_2, \dots, R_n$.
- ▶ Consider the relation **Course(course_id, title, dept_name, credits)**.

- ▶ Tuples:

course_id	title	dept_name	credits
C1	Database Systems	CS	4
C2	Algorithms	CS	3
C3	Linear Algebra	Math	3

- ▶ The relation can be reconstructed by joining:
 - ▶ **Course1(course_id, title)**
 - ▶ **Course2(course_id, dept_name)**
 - ▶ **Course3(course_id, credits)**
- ▶ on **course_id**.

Decomposition, Example 2

- ▶ Decomposition of a schema R into schemas $R_1, R_2, \dots, R_n$.
- ▶ Consider the relation **Course(course_id, hod_name, dept_name, credits)**.

▶ Tuples:

course_id	hod_name	dept_name	credits
C1	Pat	CS	4
C2	Pat	CS	3
C3	Kim	Math	3

- ▶ The relation can be reconstructed by joining:
 - ▶ **Course1(course_id, dept_name)**
 - ▶ **Course2(dept_name, hod_name)**
 - ▶ **Course3(course_id, credits)**

Aim

分解大表
↓
小表 $\rightarrow$ natural join
↓
还原
无损

- ▶ If a relation schema R can be reconstructed just by natural join from a set of schemata $R_1, R_2, \dots, R_n$ with all of them subsets of R :
- ▶ then we prefer a database with relations based on $R_1, R_2, \dots, R_n$
- ▶ we have to identify what concerns we have to take into account in this process.

Conclusion

- ▶ Joins are powerful tools in database management.
- ▶ relations sometimes can be decomposed into smaller relations and reconstructed just by natural join of all these relations.
- ▶ Embrace joins to ensure data integrity and efficiency.

Model

- ▶ In the model assumed in Relational Database Design, a relation can be constrained by business considerations in a certain way:
- ▶ In the example before the fact that the HoD name should be the same whenever a department appears is such a constraint. 约束,
- ▶ We often say, only relations are *legal* where this pattern is maintained.
- ▶ This constraint usually cannot be enforced in the database:
- ▶ This is one of the reasons why such a table experiences anomalies with any kind of change:
- ▶ Code that changes the table is more complicated than necessary, error prone and inefficient.
- ▶ This situation is considered non-optimal and the decomposition is preferred.

Update, Insert, Delete Anomalies 异常

► Update Anomaly:

- Occurs when data redundancy causes inconsistencies.
- Example: Changing the HoD name for the CS department requires updating multiple tuples. If one tuple is missed, the data becomes inconsistent.

► Insert Anomaly:

- Occurs when certain attributes cannot be inserted into the database without the presence of other attributes.
- Example: Adding a new department without any courses would be impossible if the course information is required.

► Delete Anomaly:

- Occurs when deletion of data representing certain facts necessitates deletion of data representing other facts.
- Example: Deleting the last course of a department also removes the department's HoD information.